Inference-time proofreader guidance

Can the released proofreader improve complete contact maps by changing sampling, looking ahead, maintaining several prefixes, repairing errors, or combining larger independent rollout pools?

Frozen generator: exp277 step 266344. Frozen proofreader: exp356 step 18000.
Implementation and GPU validation are in progress; no guided-policy result yet.

Frozen models and experimental references

The generator is the exp277 step-266344 Qwen3 model. The proofreader freezes its causal backbone and adds a four-layer, 512-wide bidirectional contact reader with 22.18 million trainable parameters. It predicts each supplied contact's correctness and the current rollout's recall. No model is retrained here.

The proofreader was trained on 50,000 experimental PDB chain structures. References use the complete exp222 native-only pyconfind contact set: a 3 Å side-chain threshold, minimum contact degree 0.001 and sequence separation ≥6. The validation split has 1,552 structures in 287 related groups.

What top ten percent means here

At each contact boundary, sample K candidate contacts from the constrained generator. Score them with the proofreader after appending each candidate. Keep the highest ceil(0.1 K) candidates and sample uniformly among survivors. This keeps 2 of 16 or 7 of 64 sampled proposals, including possible repeats; it does not enumerate the entire residue-pair universe.

Soft guidance weights the same proposals by q raised to a chosen power. Lookahead adds future contacts before scoring, then commits only the first. The generator's probability already enters through proposal sampling.

Complete nested-pool pilot

On 32 validation structures, sum-q R-precision increases from 0.5792 with eight rollouts to 0.5993 with 128. The paired gain is 0.0200, with 95% group-bootstrap interval 0.0050–0.0329. Long-range R-precision rises from 0.5537 to 0.5806.

Candidate coverage grows from 0.7765 to 0.9509. At 128 rollouts, frequency alone slightly exceeds sum-q: all-contact R-precision 0.6040, long-range 0.5864. Most remaining error is ranking error despite the much larger candidate pool.

Confidence tie-breaking and predicted-F1 vote weighting were added as exploratory validation tests. Their tiny gains over frequency have intervals including zero.

Baseline calibration and coverage

On 128 ordinary rollouts per pilot structure, per-rollout predicted precision has mean absolute error 0.0606 and predicted recall 0.0547. Errors average within structures and then over the same 32 structures. Contact calibration curves weight individual contacts; the two summaries answer different questions.

Mean trajectory overlap is 0.3473. Enlarging the pool exposes about 95% of native contacts, but ranking only recovers about 60% at the native contact-count budget. The remaining ranking gap motivates guidance and better aggregation; it does not establish that either can close the gap. Guided-rollout calibration is pending.

Practical single-batch baseline

Requesting each pool in one batch gives 0.6036 all-contact and 0.5894 long-range R-precision with frequency ranking at 128 rollouts. Mean warm generation cost is 7.165 H100 seconds per map; no proofreader pass is needed for this aggregator.

Against eight rollouts with sum-q ranking, the paired all-contact gain is 0.0244 (95% group interval 0.0101–0.0373) and the long-range gain is 0.0357 (0.0154–0.0519). The 128-rollout pool contains 95.39% of native contacts. The single-batch recipe is the practical cost baseline for guided search.

Larger validation confirms the pool-size benefit

On all 1,375 structures in 255 validation groups outside the pilot, frequency ranking improves from 0.5014 at eight rollouts to 0.5477 at 128. The paired gain is 0.0463 (95% group-bootstrap interval 0.0403–0.0526). Long-range R-precision rises from 0.4709 to 0.5215; its gain is 0.0506 (0.0430–0.0586).

The pool exposes 95.23% of native contacts. Mean warm generation time increases from 2.431 to 6.599 H100 seconds. Both sides use frequency ranking; fresh proofreader-weighted validation scoring remains pending. The long-range mean covers 1,371 structures with nonempty long-range references.

Benefit across lengths and measured latency

Increasing frequency-ranked pools from eight to 128 improves R-precision on 1,196 of 1,375 validation structures; 63 tie and 116 worsen. The median paired gain is 0.0392. Descriptive bands of 32–127, 128–255, 256–511 and 512–1000 residues all have positive mean all-contact and long-range gains.

For 128 rollouts, median warm generation latency is 4.345 H100 seconds, the 95th percentile is 17.024 seconds and the maximum is 63.331 seconds. Long proteins dominate the tail. Every plotted latency comes from its saved timing record; model loading and artifact writes are excluded from warm inference.

Exploratory length control

On the five short proteins completed by every original policy, top-10% filtering raises per-rollout precision but emits fewer contacts and lowers recall. This incomplete cohort motivates a stopping-rule ablation, not policy selection.

An added exploratory pilot compares ordinary and top-10%-of-16 sampling at exactly L and 2L contacts, with EOS disabled for both. Four independent 2L continuations yield nested L and 2L maps. The original 15-policy pilot remains frozen. No new reserved-test scores have been inspected.

Existing 1552-structure validation baseline

On existing validation rollouts, sum q remains strongest among five simple aggregators: all-contact R-precision 0.5315; long-range R-precision 0.5046. Frequency gives 0.5167 and 0.4877 respectively. Mean q alone gives 0.4895.

With up to eight rollouts, the pool contains about 75% of native contacts. This separates missing-contact generation from ranking errors. Oracle pool coverage is an upper bound, not the performance of an inference method.

Frozen pilot and execution

32 validation proteins from distinct homology groups span 33–937 residues. Compare top-percentile gates, soft weighting, lookahead, population search and one-pass repair, plus ordinary vLLM pools of 8, 32 and 128 rollouts.

Measure full-universe AP, all/long-range R-precision and complete search cost. Select policies on validation, then lock a reserved-test confirmation plan. Jobs use co-located CoreWeave storage and resumable per-rollout artifacts. H100 and GB200 execution phases retain separate timing provenance.

Execution and recoverability

A repeatedly preempted shared worker was canceled after repeated model staging exceeded its intended transfer budget. The incident is recorded. Recovery jobs use immutable source bundles, per-trajectory commit markers and durable S3. Regional model mirrors use persistent reservations with a 7 GB cap each.

The complete H100 ordinary baselines are saved. H100 guided and fixed-length runs were interrupted again at 18:15 UTC, and both H100 controllers stopped reporting. No shared controller, priority or other user's workload was changed. The healthy GB200 region is being evaluated as a separate hardware phase.

Numerical validation

All frozen backbone tensors match the generator exactly. CPU cache tests agree with full forward passes. The first H100 smoke caught BF16 incremental/full-pass drift of up to 0.02394 in one contact probability. FP32 cached search passes on three real proteins, with maximum error 0.00000358. Final maps use the released BF16 full-pass reader for comparison.

GB200 verification and its limits

Tiny-model cache checks and production ordinary generation passed on GB200. The production proofreader's CUDA 13 replay failed: the largest BF16 contact-score difference from saved H100 predictions was 0.0800, against a 0.01 guard. A real prefix also exceeded the unchanged 1e-5 FP32 cache-equivalence threshold.

Forcing math attention did not meet the BF16 guard; memory-efficient attention was unsupported for this cache path. Matching CUDA 12.8 passed the BF16 replay: two batches match exactly and the third has maximum difference 0.001523. One cache test still fails, so cached guidance remains pending. Ordinary generation and full-forward scoring can proceed. Cost plots reject mixtures of GPU models.

How to interpret the eventual comparison

R-precision evaluates the top R predictions, where R is the number of native contacts used only for evaluation. Precision at L, L/2 and L/5 provides usable sequence-length budgets. AP includes every eligible unproposed pair at score zero. Contact labels enter only after sampling decisions are finished.

Pilot and larger-validation results select the final methods. The reserved test was previously used for the proofreader release; new inference-policy test scores have not been inspected. Generator pretraining decontamination is not established. Search timings describe this implementation, including discarded work, and are not hardware-independent lower bounds on the cost of these policy families.

Stronger comparisons at larger compute budgets

A separate GB200 ordinary-generation phase requests 512 and 2,048 independent rollouts per pilot structure. These sizes were frozen before complete guided results were available. Observed lookahead costs already greatly exceeded the 128-rollout baseline, motivating a wider ordinary-sampling cost curve.

Frequency ranking needs no proofreader pass. These are measured compute envelopes, not exact budget matching. The same 32 validation structures are used, and the reserved test remains untouched for new policy metrics.

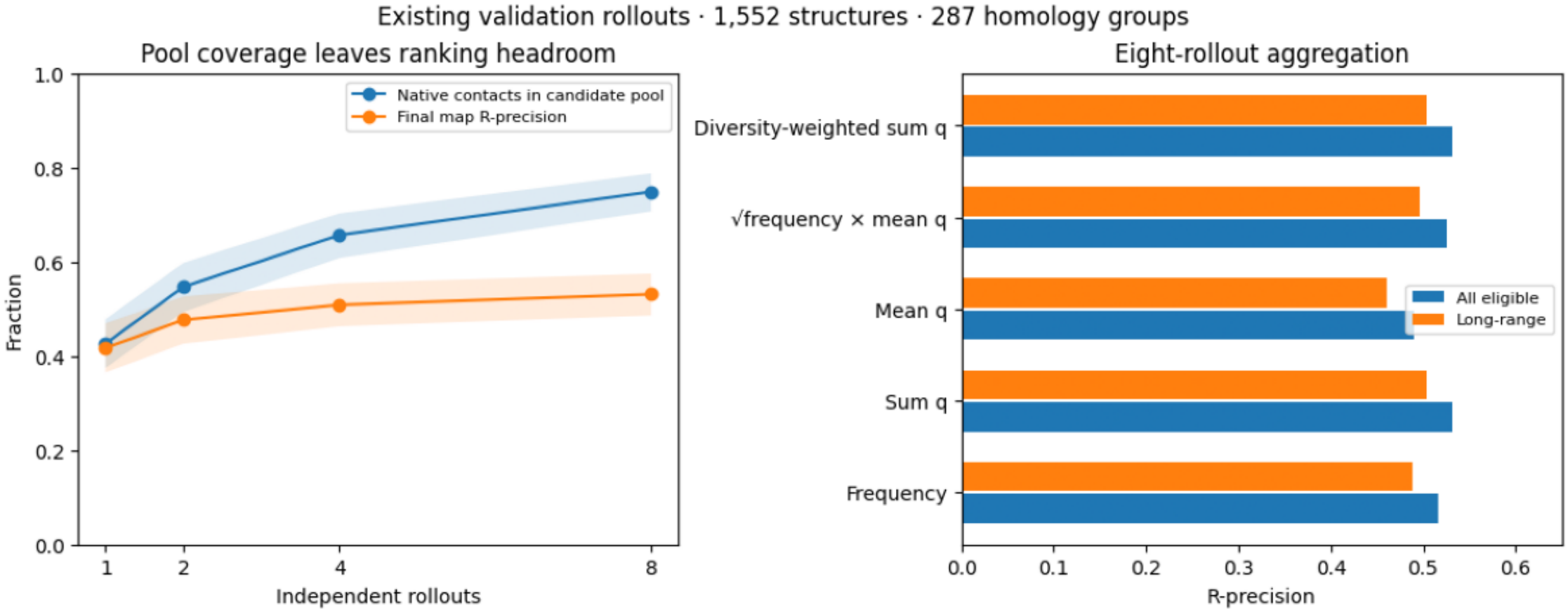
Reranking helps the small pilot pools

Replacing frequency by sum q changes 15.37% of top-R slots with eight rollouts and improves R-precision by 0.0163 (paired 95% interval 0.0090–0.0248). With 128 rollouts, it changes 7.62% of slots; the net effect is −0.0031 (−0.0099–0.0039), which does not establish a difference.

The diagnostic counts true contacts entering and leaving the map. Those counts exactly reproduce the observed metric difference for every protein. All 32 pilot structures are included; these results do not replace the larger validation test.

existing_validation.png

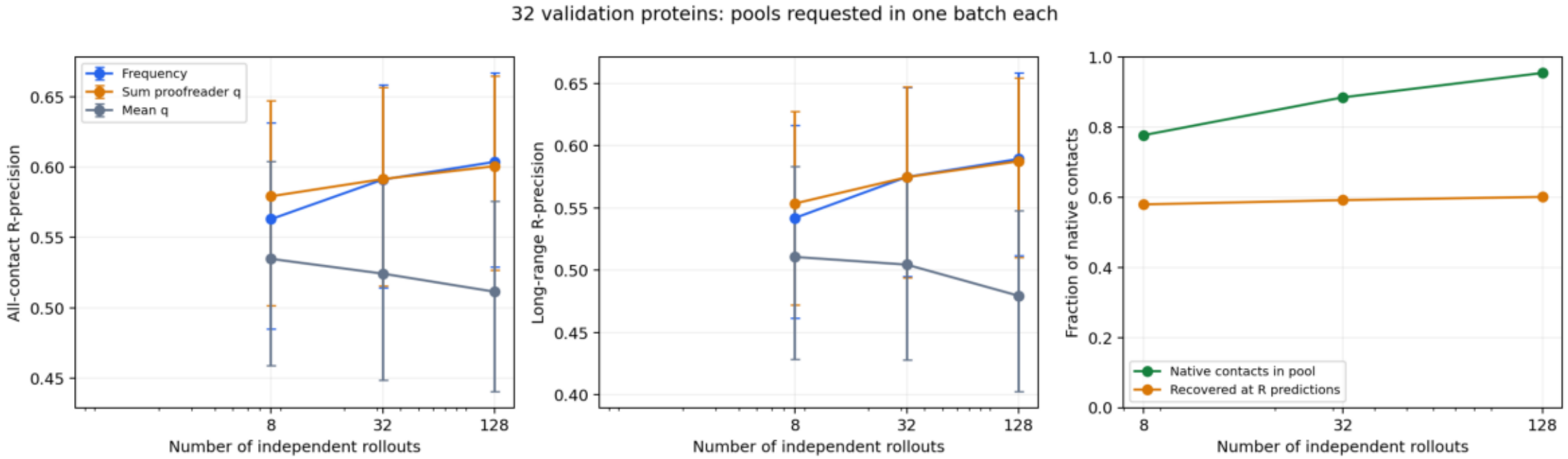
Validation-only reanalysis. Bands resample homology groups (95% intervals). Oracle coverage uses labels only to bound achievable recall. The small number of structures with fewer than eight usable rollouts retains all available samples. Long-range means separation ≥24.



$ plot_existing.py

pilot_direct_pool.png

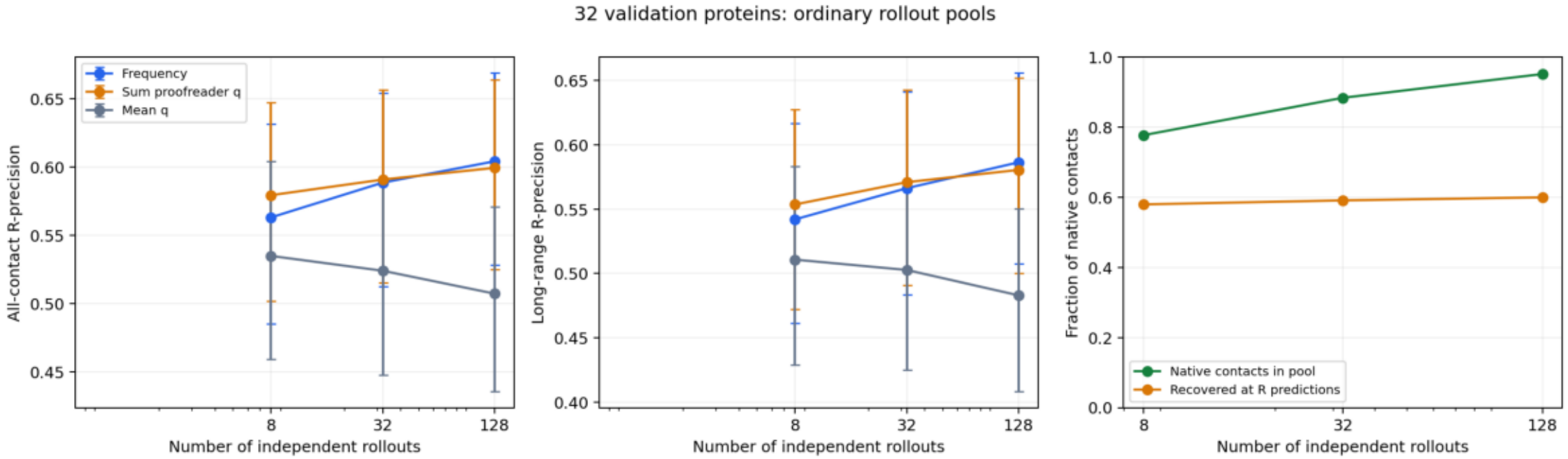
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_direct_pool.png --title '32 validation proteins: pools requested in one batch each'
```

pilot_pool.png

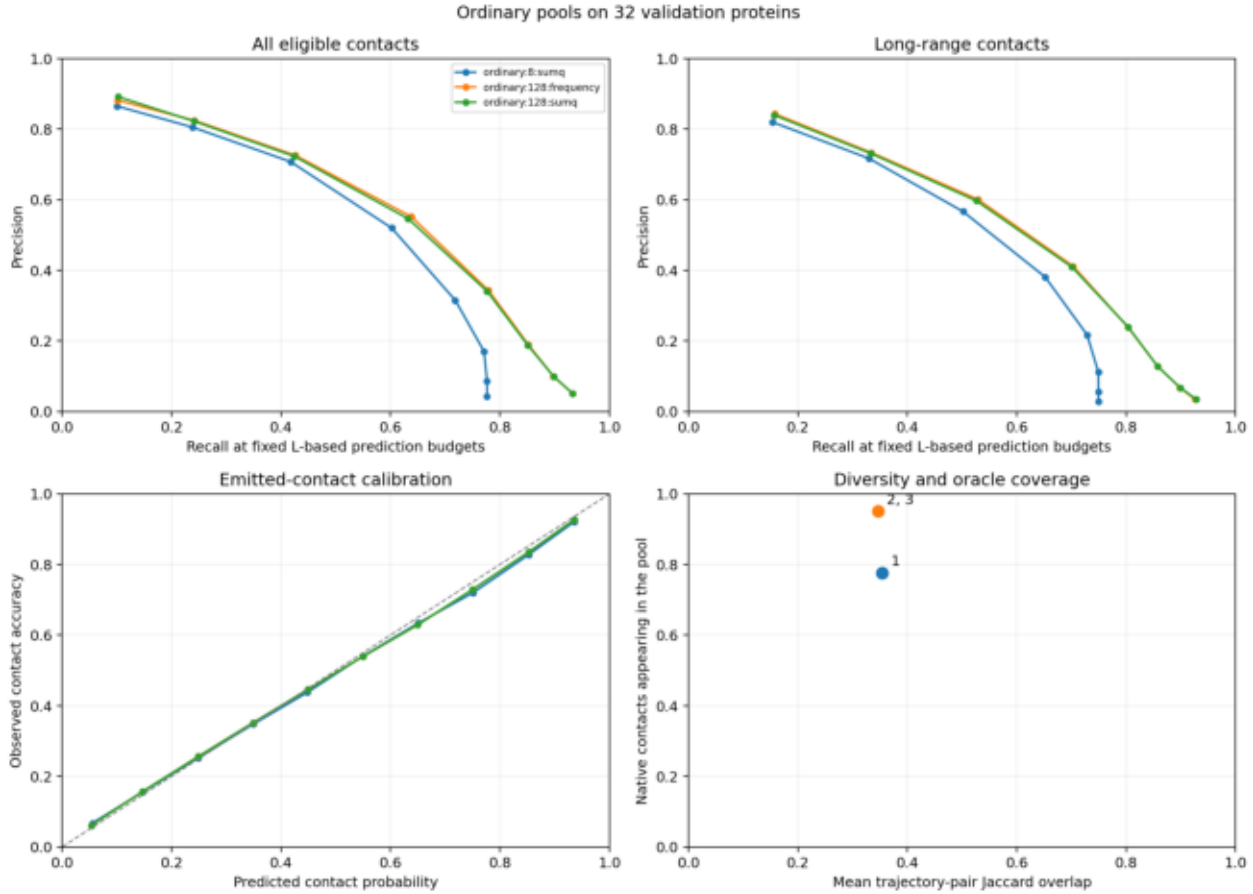
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool.png
```

pilot_pool_diagnostics.png

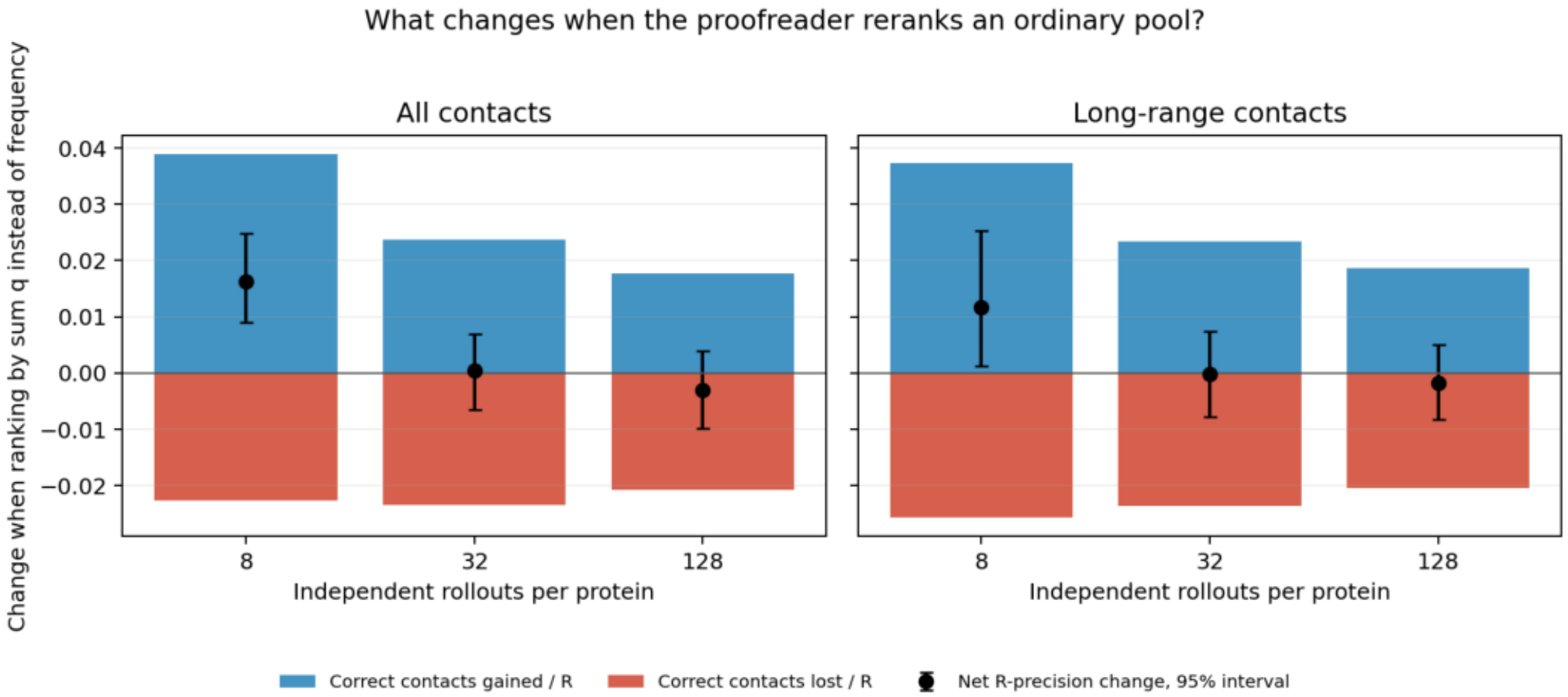
Curves average over the same structures at budgets 0.1, 0.25, 0.5, 1, 2, 4, 8 and 16 times sequence length. Unfilled slots count as misses; curves do not use native contact counts to choose the prediction budget. Calibration pools first occurrences within each rollout and is contact-weighted. Diversity and coverage are structure means; at most 128 deterministically sampled trajectory pairs measure overlap. Numbers in the coverage panel follow legend order; multiple aggregators of the same pool share calibration and diversity.




```
$ plot_diagnostics.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool/diagnostics --spec ordinary:8:sumq ordinary:128:frequency ordinary:128:sumq \
  --title 'Ordinary pools on 32 validation proteins' --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool_diagnostics.png
```

pilot_reranking.png

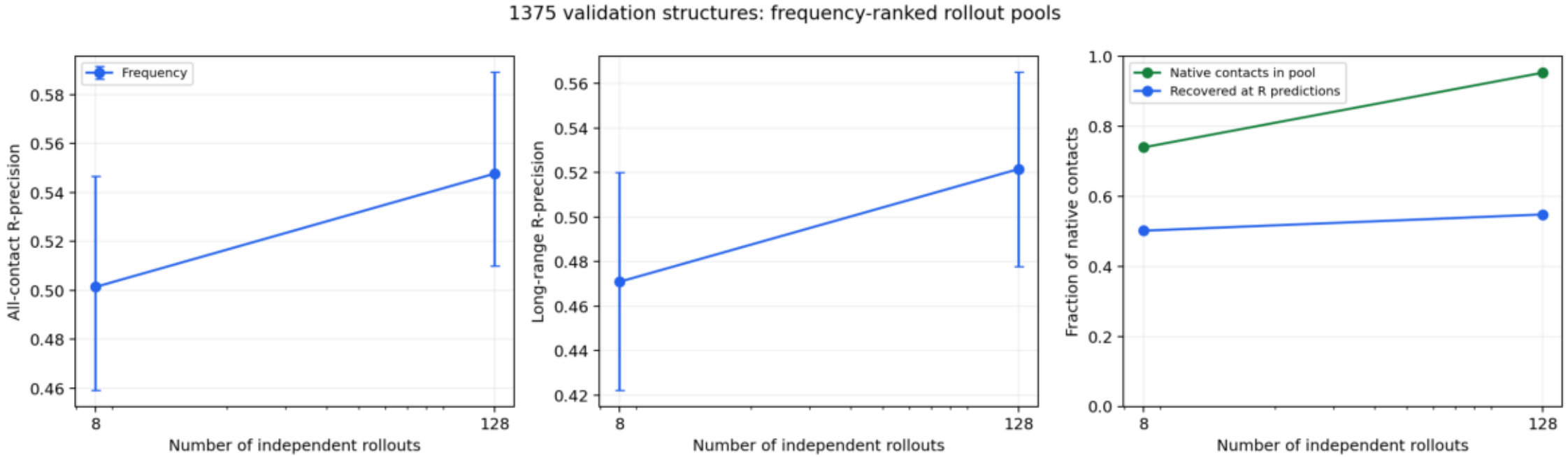
Both rankings use exactly the same sampled candidate union and deterministic label-independent tie breaks. Bars count true contacts entering and leaving the top R; their sum equals the observed R-precision change, verified for every input. Proteins are weighted equally; intervals resample homology groups. R uses native contact count only for retrospective evaluation. This is an adaptive diagnostic, not a new policy.




```
$ analyze_reranking.py --root experiments/exp368_evals_proofreader_guided_contact_maps/_cache/pilot-direct-scored --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/pilot_direct.json --targets experiments/exp356_models_pdb_rollout_proofreader/_cache/prepared-v2/targets.parquet --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --plot experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_reranking.png
```

validation_frequency.png

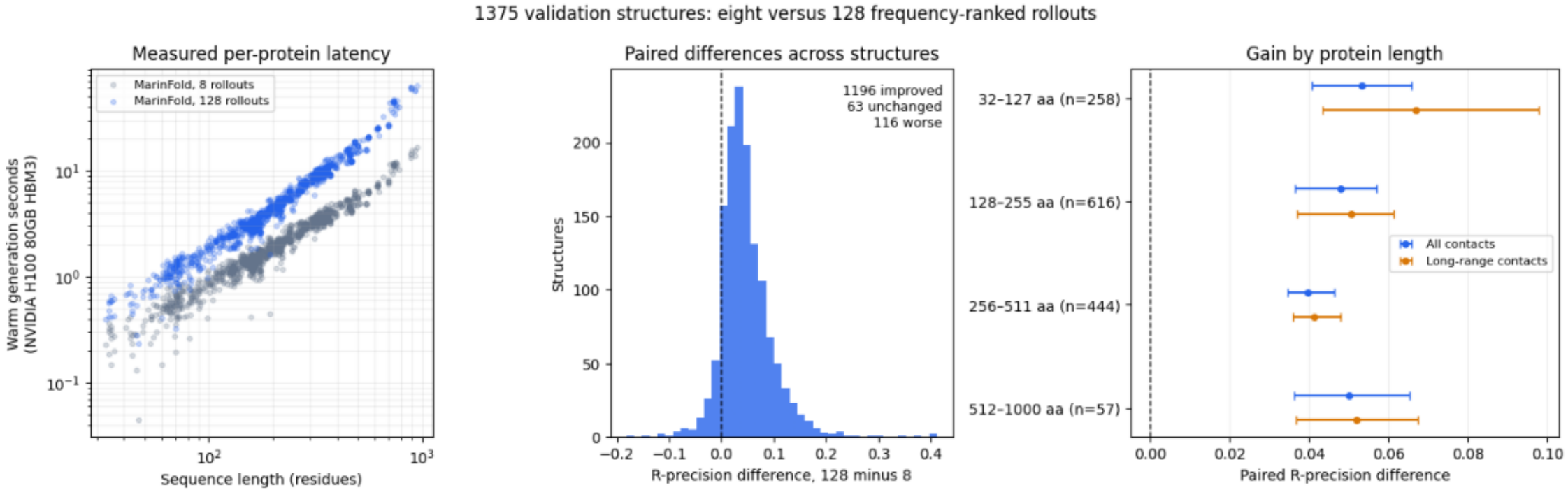
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_frequency.png --title '1375 validation structures: frequency-ranked rollout pools' --methods frequency \
  --coverage-method frequency
```

validation_pool_scaling.png

Each latency point is one (protein, pool-size) inference, excluding model load and artifact writes. Length strata use fixed boundaries at 32, 128, 256, 512 and 1001 residues. Intervals resample related groups within each stratum and describe paired differences. Strata are descriptive; they do not select proteins or tune a test policy.




```
$ plot_pool_scaling.py --input experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency/per_protein.csv.gz --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/validation_pool.json --data-out experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_pool_scaling.png
```